Development and application of a dynamic flame extinction and re-ignition model in FDS

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Abstract

A new dynamic flame extinction and re-ignition model that considers the local properties of the flow has been developed. The extinction model is based on the default enthalpy balance used by the Fire Dynamics Simulator (FDS) and employs a dynamic calculation of the critical flame temperature, T_{CFT}, derived from well-stirred reactor simulations with different diluent concentrations, which depends on the local mixing time and species concentrations. On the other hand, re-ignition is allowed to occur if the cell temperature is above a threshold re-ignition temperature, T_{ign}, derived from counterflow simulations considering hot fuel and oxidizers, which depends on the local strain rate magnitude. The model has been implemented in FDS and validated against two well-characterized scenarios with different fuels (i.e., UMD line burner and FM burner cases) involving flame extinction due to dilution. Overall, improved predictions with respect to the reduction of the combustion efficiency, when compared to the original extinction and re-ignition model of FDS, are obtained. The predicted T_{ign} and T_{CFT} were (significantly) higher and lower, respectively, compared to the default value of FDS. In addition, lower O_2 concentrations were qualitatively observed within the flame region with the new model.

Keywords: flame extinction, flame re-ignition, critical flame temperature, strain rate, CFD modelling, FDS

1. Introduction

The Computational Fluid Dynamics (CFD) code Fire Dynamics Simulator (FDS) [1], which has been developed by the National Institute of Standards and Technology (NIST), is adopted in fire safety engineering practice more than ever due to its continuous improvement, development and validation as well as due to the increasing accessibility to high-performance computing facilities. However, simulations involving flame extinction and ventilation-controlled compartment fires still remain challenging [2, 3, 4]. This may be attributed to possible limitations of the employed extinction and re-ignition sub-models, and the mesh dependence thereof. Engineers that use FDS for design and engineering purposes rarely challenge the accuracy of CFD results as they usually carry out blind simulations without full consideration of model uncertainties. However, ignoring such potential inaccuracies in specific scenarios may lead to important limitations and unsupported conclusions. Therefore, it becomes fundamental to understand the accuracy and uncertainties of the CFD models and enhance them to extend their range of applicability towards accurately modelling ventilation-limited combustion regimes.

In the context of fire modelling using CFD, flame extinction is often modelled by using either a critical flame temperature (CFT) approach (e.g., [5, 6, 7]) or a critical Damköhler number formulation (e.g., [8, 9, 10, 11, 12]) among others. The CFT method assumes that a flame is extinguished once the local temperature falls below a specified threshold. This threshold is fuel dependent but is often treated as a constant, typically ranging between 1500 - 1800 K [13]. On the other hand, flame (re-)ignition in these models is only allowed if the local gas temperature in a cell is above a constant threshold temperature (i.e., set either to the auto-ignition temperature (AIT) of the fuel or to a generic constant value, e.g., on the order of 1000 K [9]).

Since flame extinction and re-ignition of quenched fuel are kinetically controlled phenomena, the most comprehensive modelling strategy would require incorporation of the finite-rate chemistry mechanisms in the large eddy simulations of fire development. This strategy was implemented in recent numerical studies [14, 15] where the Eddy Dissipation Concept (EDC) was applied. Although potentially capable of predicting both extinction and re-ignition without additional sub-models, this approach was found to be very sensitive to the selection of a chemical mechanism. Practical applicability of EDC is also limited by the inherent assumption of fully developed turbulence used in its derivation. To overcome these limitations, an alternative sub-grid combustion model (SCM) incorporating

the finite-rate chemistry of fuel oxidation has been developed in [16] and validated for unconfined flames in [17, 18]. The finite-rate chemistry models are more computationally expensive and still require further validation for under-ventilated enclosure fires.

The main goal of this research study concerns improving the accuracy and applicability of FDS in scenarios involving flame extinction and vitiated conditions. As such, improvements to the extinction and re-ignition models currently available in FDS are proposed. To this purpose, instead of prescribing constant values as done by default by FDS, a dynamic calculation, considering the local flow conditions, of the model parameters related to extinction and re-ignition is considered. Subsequently, the developed models are implemented in FDS and their accuracy is evaluated against well-documented experiments, which are part of the MaCFP workshop [19], involving flame extinction due to dilution. Comparisons against the original extinction and re-ignition models of FDS are also made.

2. Current FDS models

2.1. Extinction model

The original extinction model of FDS in Large Eddy Simulation (LES) mode considers an enthalpy balance based on the concept of a critical flame temperature. The criterion examines whether a reactant mixture within a cell will have enough energy to raise its temperature above a critical flame temperature to sustain combustion [20]:

$$Z_F^0 h_F(\widetilde{T}) + \widetilde{\phi} Z_A^0 h_A(\widetilde{T}) + \widetilde{\phi} Z_P^0 h_P(\widetilde{T}) < Z_F h_F(T_{CFT}) + [Z_P - (1-\widetilde{\phi}) Z_P^0] h_P(T_{CFT}) \tag{1}$$

where Z_F^0, Z_A^0, Z_P^0 and Z_F, Z_A, Z_0 are the mass fractions of lumped species fuel, air, and products in the mixed portion of the cell at the start and the end of the reaction step, respectively, $\widetilde{T}$ is the initial cell temperature, T_{CFT} is the critical flame temperature, while $h_F(\widetilde{T})$ and $h_P(\widetilde{T})$ is the sensible enthalpy of fuel and products, respectively. Here $h_A(\widetilde{T})$ accounts for both the chemical and sensible enthalpy; thus, the left-hand-side of Eq. (1) includes the heat release due to combustion. The products include the products of combustion and any diluents.

A modified form of the equivalence ratio, $\widetilde{\phi}$, is defined as:

$$\widetilde{\phi} \equiv \left(1, \frac{s Z_F^0}{Z_A^0}\right) = \frac{Z_A^0 - Z_A}{Z_A^0} \tag{2}$$

where s is the mass stoichiometric coefficient for air.

Based on the model formulation, any excess fuel within a cell acts as a diluent while any excess air and a proportional amount of products do not. To achieve this, a fraction of the total mass of the cell equal to $(1-\widetilde{\phi})(Z_A^0 + Z_P^0)$ has been removed from the enthalpy calculation. The right-hand side of Eq. (1) contains no air because all excess air has been removed from the calculation. In other words, for an infinitely fast reaction it is $Z_A - (1-\widetilde{\phi})Z_A^0 \equiv 0$. If the inequality is true then reaction within a cell is not allowed during the current time step. For the evaluation of Eq. (1), a constant critical flame temperature, T_{CFT}, which is fuel-dependent and in the range of 1500 - 1800 K [13], is used by FDS.

2.2. Re-ignition model

If local extinction occurs, unburned fuel gas may re-ignite if it later mixes with enough oxygen elsewhere in the domain. However, this re-ignition might not be physically realistic due to insufficiently low temperatures to sustain combustion. To prevent such spurious re-ignition phenomena, re-ignition is only allowed if the cell temperature, $\widetilde{T}$, is above a threshold re-ignition temperature, T_{ign} [20]:

$$\widetilde{T} \geq T_{ign} \tag{3}$$

where T_{ign} is a constant and set to the Auto-Ignition Temperature (AIT) of the fuel.

The AIT serves to stop fuel and oxygen from automatically re-igniting. Nevertheless, to ensure the primary combustion zone is not suppressed, a small exclusion region (typically a few cells surrounding the burner) is defined where combustion cannot be suppressed. Ideally, the size of the exclusion zone needs to be large enough so as not to affect the primary source of combustion (i.e., to avoid full extinction and provide a stabilized flame) but also small enough so that the (re-)ignition and extinction models can be evaluated in a relatively large portion of the flame (i.e., where most chemical reactions take place).

3. Development of dynamic models

The modelling approaches previously explored in [21]-[22], which use local flow properties for a dynamic estimation of the critical flame extinction temperatures, T_{CFT}, and re-ignition temperatures, T_{ign}, are used as a starting point for model development. In any fire scenario, it is expected that the flame behavior will be influenced by the local strain rate by changing the residence time available for the mixture to react. As such, it is reasonable to assume that re-ignition of any

unburned fuel can only occur at temperatures higher than the AIT of the fuel (i.e., the default approach of FDS). Similarly, local strain effects will potentially also vary the critical flame temperature needed for a sustained flame. As such, the re-ignition temperatures will be estimated on the basis of counterflow simulations of different fuels against hot oxidizers at different strain rates. On the other hand, the determination of the critical flame temperatures will be made through well-stirred reactor simulations under different residence times and diluents.

3.1. Determination of re-ignition temperatures

The counterflow diffusion flame model by Cantera [23] is used for the 1D simulations of an axisymmetric stagnation flow. The configuration considered in the numerical simulations is shown graphically in Fig. 1.

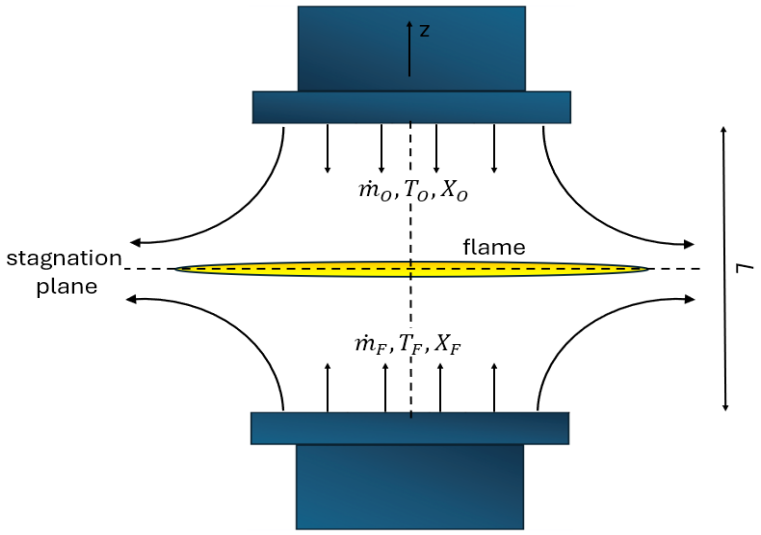


Figure 1: Schematic of the 1D counterflow flame configuration. Variables $\dot{m}$, T and X denote the mass flow rate per unit area, temperature and mole fraction, respectively, while the subscripts F and O denote the fuel and oxidizer, respectively.

The three-dimensional governing equations are reduced to a single dimension using a similarity solution [24]. Species transport is modelled using Cantera's multi-component formulation based on the model described by Dixon-Lewis [25]. This model solves the full system of Stefan-Maxwell equations, thereby accounting for cross-diffusion effects. To improve fidelity, the Soret effect (thermal diffusion) is also included in the model. The radiation model follows the approach of Liu and Rogg [26] and it relies on the optically thin limit together with the

gray-gas approximation, which allows a simplified evaluation of the volumetric heat flux. In this framework, gas emission is calculated while absorption and re-emission within the medium are neglected. The heat flux is obtained from Planck absorption coefficients, boundary emissivities, and temperature. Polynomial correlations are used to determine the Planck coefficients of CO_2 and H_2O, with input data sourced from RADCAL [27], and the polynomial coefficients derived from [28]. Any contribution to radiative heat transfer by soot is ignored. Adaptive mesh refinement was utilized along the centerline to resolve sharp gradients within the flame front, with grid insertion triggered when local relative slopes exceeded 2% or curvatures exceeded 5%, while the maximum ratio between adjacent grid intervals was limited at 2 to maintain mesh smoothness. Redundant grid points were dynamically removed where relative variations fell below 0.2%, ensuring a highly resolved solution.

3.1.1. Counterflow flame simulations

Four different fuels, i.e., methane (CH_4), ethane (C_2H_6), propane (C_3H_8), and butane (C_4H_{10}) are considered in the current work based on the available experimental data reported in [29, 30]. The main required user-input data for the counterflow simulations include the composition and temperature for the fuel and oxidizer, the separation distance between fuel and oxidizer and the global strain rate. As such, cold (i.e., 300 K) diluted fuel-air mixtures (i.e., 8% CH_4 in N_2, 6% C_2H_6 in N_2, 6% C_3H_8 in N_2 and 5% C_4H_{10} in N_2), in the fuel stream, versus heated air at ambient oxygen concentration, in the oxidizer stream, at a pressure of 1 atm are considered. The separation distance between the two streams is $L = 2$ cm with the momentum fluxes of both streams set to be equal. The oxidizer temperatures in the simulations are set to vary between 600 - 1500 K, with 25 K increments, until successful ignition of the fuel occurs.

The global strain rate, defined as $S_g = (u_F + u_O)/L$ where u is the velocity, is used for characterizing the inflow conditions of the two streams. The considered global strain rates varied between 5 - 600 s^{-1}. To be consistent when comparing against the experiments, however, the results are plotted as a function of the flow strain rate, determined based on the maximum velocity gradient between the two streams defined as $S = -(du/dz)_{max}$.

An overview of the reaction mechanisms considered in the counterflow flame simulations is presented in Table 1. The chosen mechanisms have been validated across a range of fuels and flame configurations.

Table 1: Overview of the reaction mechanisms considered in the counterflow simulations.

Mechanism	Fuel	Species	Reactions
San Diego [31]	C_0-C_4	57	268
USC II [32]	C_0-C_4	111	784
FFCM-2 [33]	C_0-C_4	96	1054
CRECK [34]	C_1-C_3	114	1999
GRI 3.0 [35]	C_1	53	325
Wang [36]	C_2-C_3	75	529
Peters [37]	C_3	31	107

3.1.2. Results

A comparison between the predicted re-ignition temperatures from the simulations and the experimental data is presented in Figure 2. The reported AIT [13] of fuels and a 5% experimental uncertainty are also included in the figures.

Across all fuels, the re-ignition temperatures are close to the AIT of the fuel at very low strain rates and increase sharply between 5 - 150 s^{-1} before leveling off beyond approximately 200 - 300 s^{-1}, indicating a transition to strain-insensitive re-ignition behavior. The different mechanisms show similar qualitative trends, though quantitative differences are evident, particularly at lower strain rates. The overall trend shows that heavier hydrocarbons (C_3H_8 and C_4H_{10}) ignite at slightly lower temperatures than lighter ones (CH_4 and C_2H_6) under the same strain rate conditions. While the various mechanisms capture the general trends, notable discrepancies can be identified depending on the fuel, e.g., for CH_4 the CRECK and GRI 3.0 mechanisms tend to somewhat over- and under-predict the ignition temperatures, respectively, in the region where experimental data are available. Similar observations can be made by examining the other fuels and reaction mechanisms as well. Nevertheless, it is worth noting that no available experimental data exist for strain rates lower than 100 s^{-1}, a region which is of interest for fire applications. As such, some uncertainty regarding the accuracy of the mechanism in this region will inevitably exist. Overall, the FFCM-2 mechanism provided predictions in closer agreement to the experimental data, followed by predictions of USC II. As such, the FFCM-2 mechanism was chosen for the remaining simulations in this study. Further validation of the accuracy of the FFCM-2 mechanism at low strain rates can be considered in the future when experimental data become available.

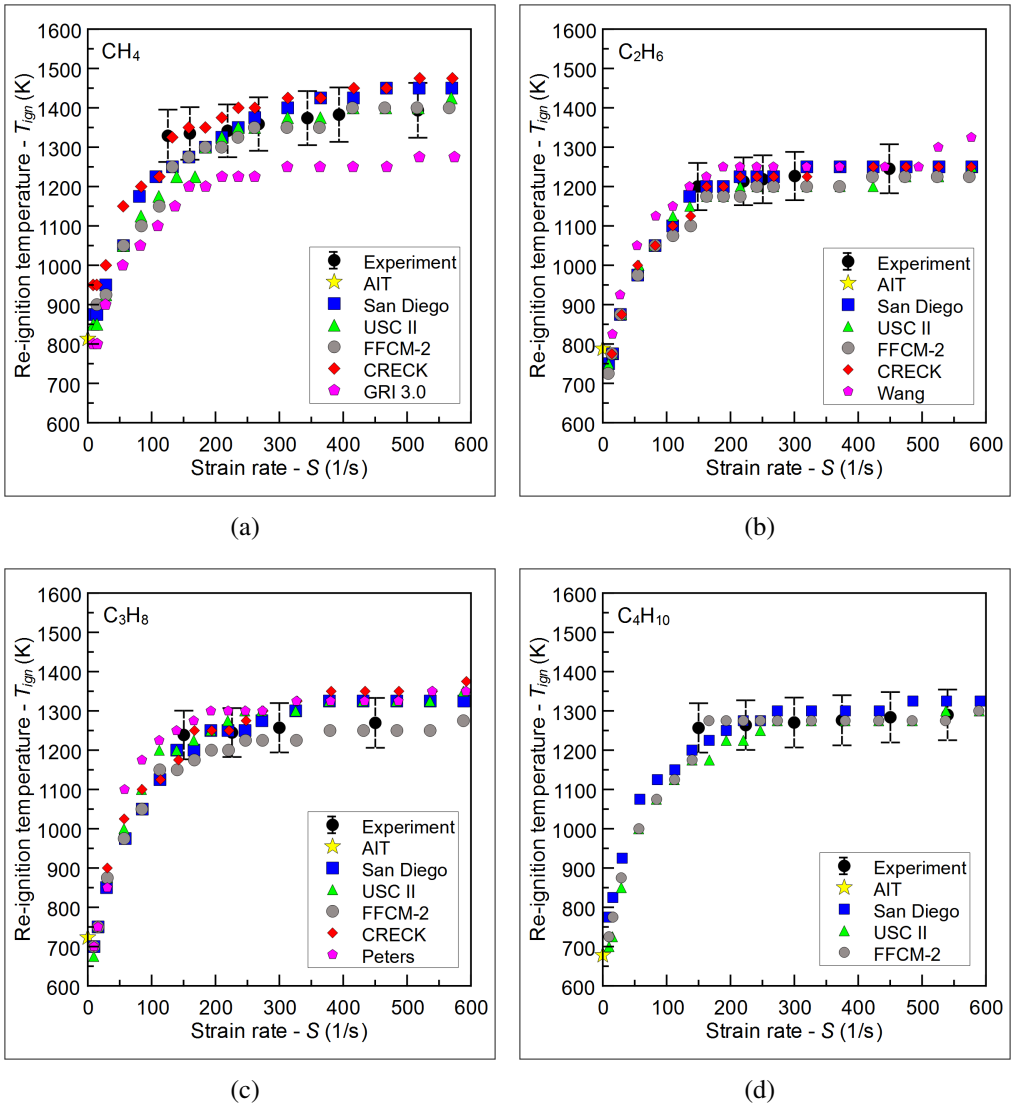


Figure 2: Predicted re-ignition temperatures for (a) CH_4, (b) C_2H_6, (c) C_3H_8 and (d) C_4H_{10} with different reaction mechanisms in the counterflow simulations. The AIT values for CH_4, C_2H_6, C_3H_8 and C_4H_{10} are 813 K, 788 K, 723 K and 678 K [13], respectively.

Figure 3 shows the influence of the ambient oxygen concentration on the predicted re-ignition temperatures of CH_4, C_2H_6, C_3H_8 and C_4H_{10} using the FFCM-2 mechanism and a cold fuel (300 K). Overall, the ambient oxygen content only has a minor effect on the ignition behavior, with re-ignition temperature differences

between 21%, 18%, 15%, and 12% O_2 becoming noticeable mainly at low strain rates, near the extinction limit. This sensitivity is mainly pronounced for methane where lower ambient oxygen concentrations significantly increase the re-ignition temperature for 12% O_2. For the heavier hydrocarbons, the dependence on the ambient O_2 concentration is weaker, and comparable re-ignition temperatures are obtained as the strain rate increases, indicating that at higher strain rates, re-ignition is primarily controlled by flow dynamics rather than by oxidizer concentration. Given the fact that the effect of vitiation is not as strongly observed in the results for all fuels, the influence of the ambient O_2 concentrations is not included in the determination of the re-ignition temperatures (i.e., only the effect of strain rate is considered).

In practical fire scenarios, flammable fuel vapors are often released at elevated temperatures, i.e., higher than ambient. Also, within a CFD cell there is only a single temperature representative of the mixture (i.e., fuel and oxidizer). As such, the derivation of re-ignition temperatures considering both the fuel and oxidizer to be hot (and at the same temperature) is physically more appropriate and more consistent with the CFD modelling. Therefore, an additional series of simulations have been performed, with the exact same settings as previously but with a fuel temperature set equal to the oxidizer temperature. These are labelled 'hot fuel'. The predicted re-ignition temperatures with the FFCM-2 mechanism are presented in Figure 4. For comparison purposes, the experimental data and numerical results considering a fuel temperature of 300 K ('cold fuel'), previously shown in Figure 2, are also included here. As expected, the predicted re-ignition temperatures considering a hot fuel are significantly lower compared to a cold fuel, with increasing differences as the strain rate increases.

The numerical data from the counterflow simulations with the FFCM-2 mechanism considering a hot fuel for the different fuels (shown in Figure 4) are then fitted using a logarithmic function in the following form:

$$T_{ign} = a\ln(S) + b \tag{4}$$

where S (1/s) is the strain rate while a and b are two model parameters which are fuel-dependent. In the CFD simulations, the resolved strain rate magnitude in every cell will be used for estimating the re-ignition temperatures, T_{ign}.

The re-ignition model parameters, the R-squared coefficient and the range inside which the re-ignition temperatures are clipped are reported in Table 2. The considered clipping range is based on the minimum and maximum re-ignition temperatures predicted from the counterflow simulations.

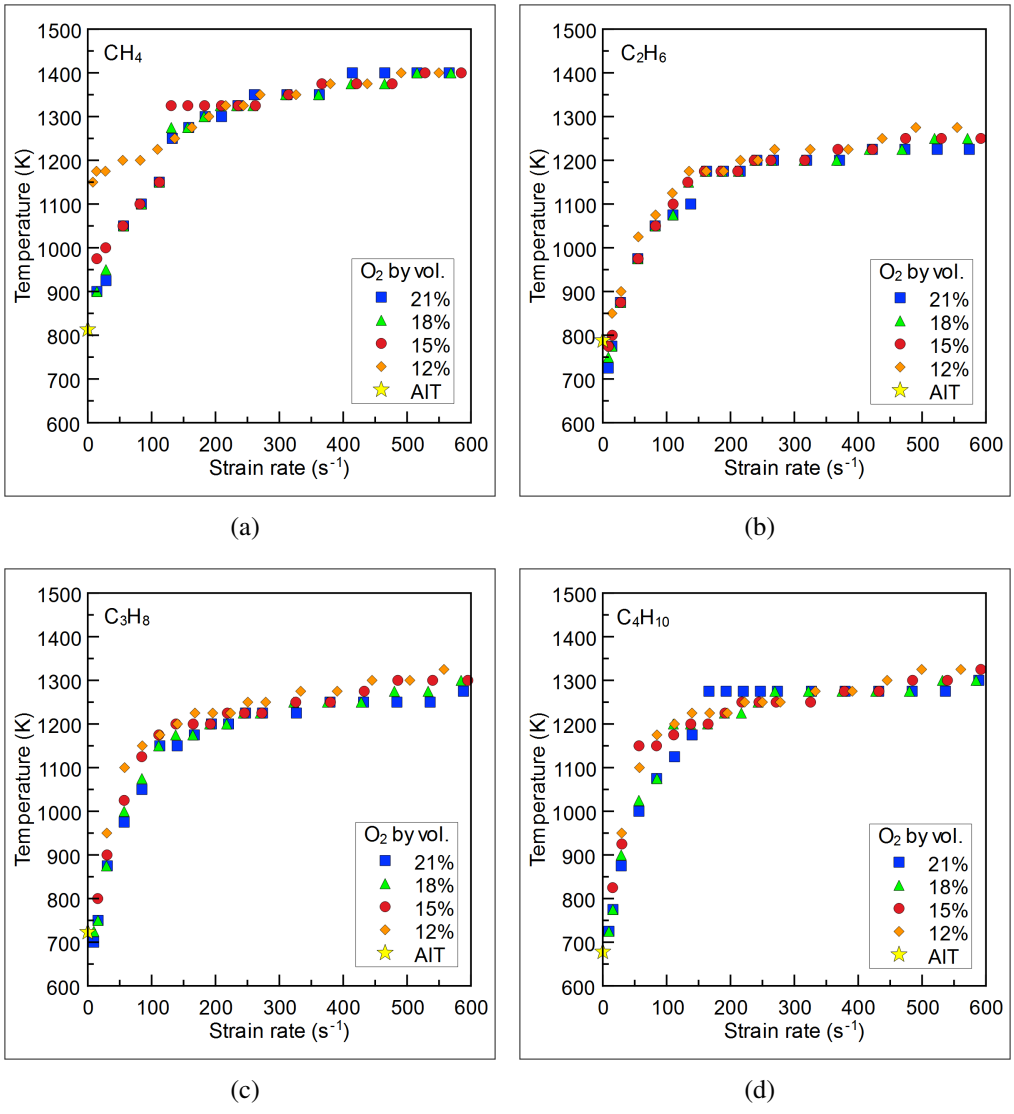


Figure 3: Predicted ignition temperatures with the FFCM-2 reaction mechanisms and different ambient O_2 (% vol.) concentrations for (a) CH_4, (b) C_2H_6, (c) C_3H_8 and (d) C_4H_{10} in the counterflow simulations with cold fuel (300 K).

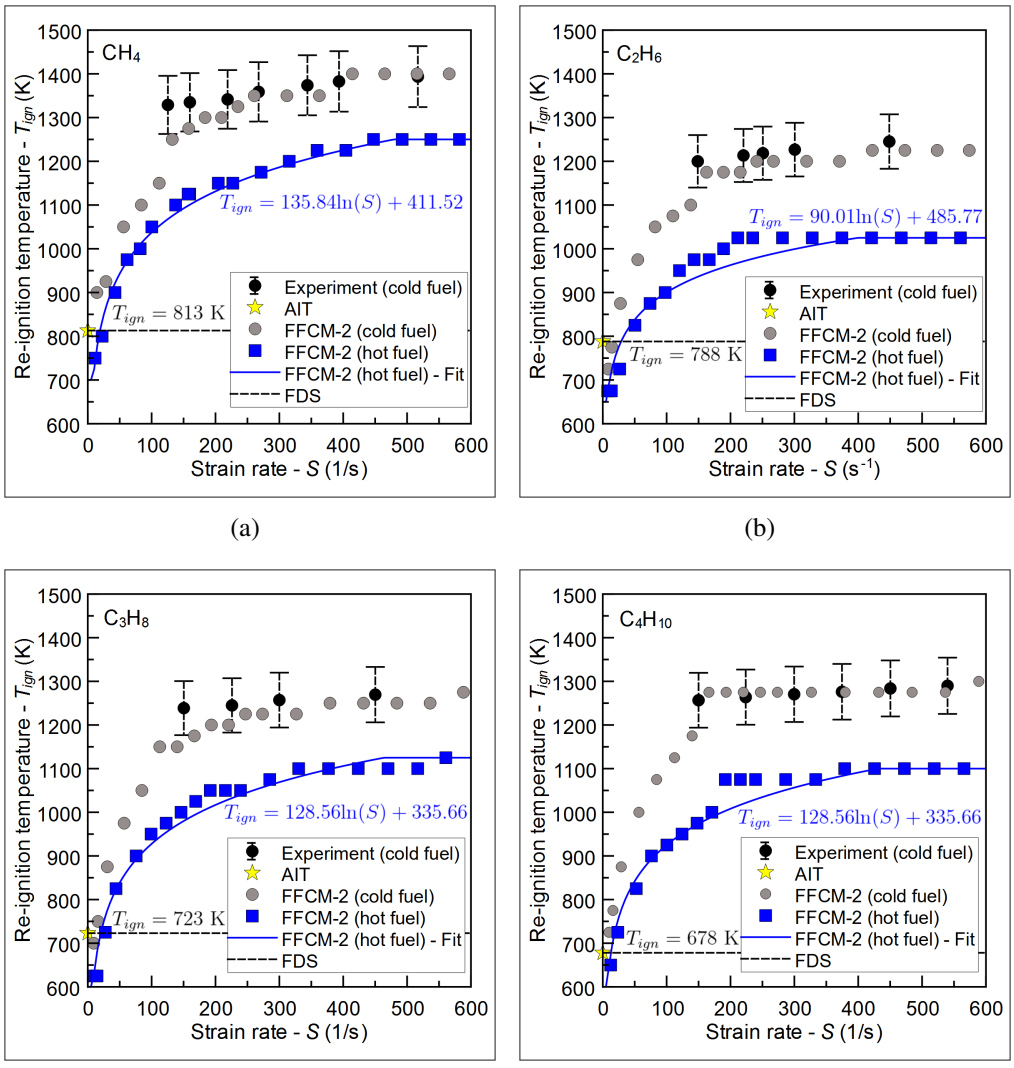


Figure 4: Predicted re-ignition temperatures with the FFCM-2 reaction mechanism and different fuel temperatures for (a) CH_4, (b) C_2H_6, (c) C_3H_8 and (d) C_4H_{10} in the counterflow simulations. 'Cold fuel' implies a fixed fuel temperature at 300 K. 'Hot fuel' means that the temperature of the fuel is equal to the temperature of the oxidizer.

Table 2: Model parameters used in the re-ignition model with different fuels.

Fuel	a	b	R^2	Clipping range
$\mathbf{CH_4}$	135.84	411.52	0.99	700 - 1250 K
$\mathbf{C_2H_6}$	90.01	485.77	0.91	650 - 1025 K
$\mathbf{C_3H_8}$	128.56	335.66	0.97	600 - 1125 K
$\mathbf{C_4H_{10}}$	119.68	373.94	0.95	600 - 1100 K

3.2. Determination of critical flame temperatures

3.2.1. Well-stirred reactor simulations

The simulations are performed using a perfectly stirred reactor (WSR) model in Cantera [38], employing the FFCM-2 reaction mechanism and considering the same fuels as previously (i.e., CH_4, C_2H_6, C_3H_8 and C_4H_{10}). The reactor pressure is fixed at 1 atm, while the inlet temperature is set to 300 K and the reactor is initialized at 1300 K to promote ignition. The reactor volume is set to 1×10^{-3} m^3, with inlet and outlet represented as reservoirs connected through a mass flow controller to impose the desired residence time and a pressure controller to maintain constant pressure. Residence times are varied between 10^{-5} and 1 s using 100 discrete steps. Following [39], various inlet compositions are considered to represent different levels of dilution with N_2, CO_2, and H_2O, ranging from 0% to 50 - 60%, while maintaining stoichiometric oxidizer-to-fuel ratios. Radiation heat losses are included through a Stefan-Boltzmann wall model assuming an emissivity of 0.8, an ambient temperature of 300 K, and a surface area estimated from the reactor volume. Steady-state convergence is determined based on a temperature tolerance of 10^{-8} K between successive time steps. The resulting steady-state reactor temperatures are recorded as a function of the residence time.

3.2.2. Results

Figure 5 presents the steady-state temperatures as a function of residence time for the different fuels and for different types/amounts of diluents (i.e., N_2, CO_2 and H_2O). As expected, a decrease in the predicted steady-state reactor temperature is observed with both decreasing residence time (i.e., time the reactants stay inside the reactor) and increasing amount (vol. $\%$) of diluent. Eventually, and for a given residence time, the mixture goes to extinction. For clarity, not all the different dilution $\%$ considered in the simulations are shown in the figures.

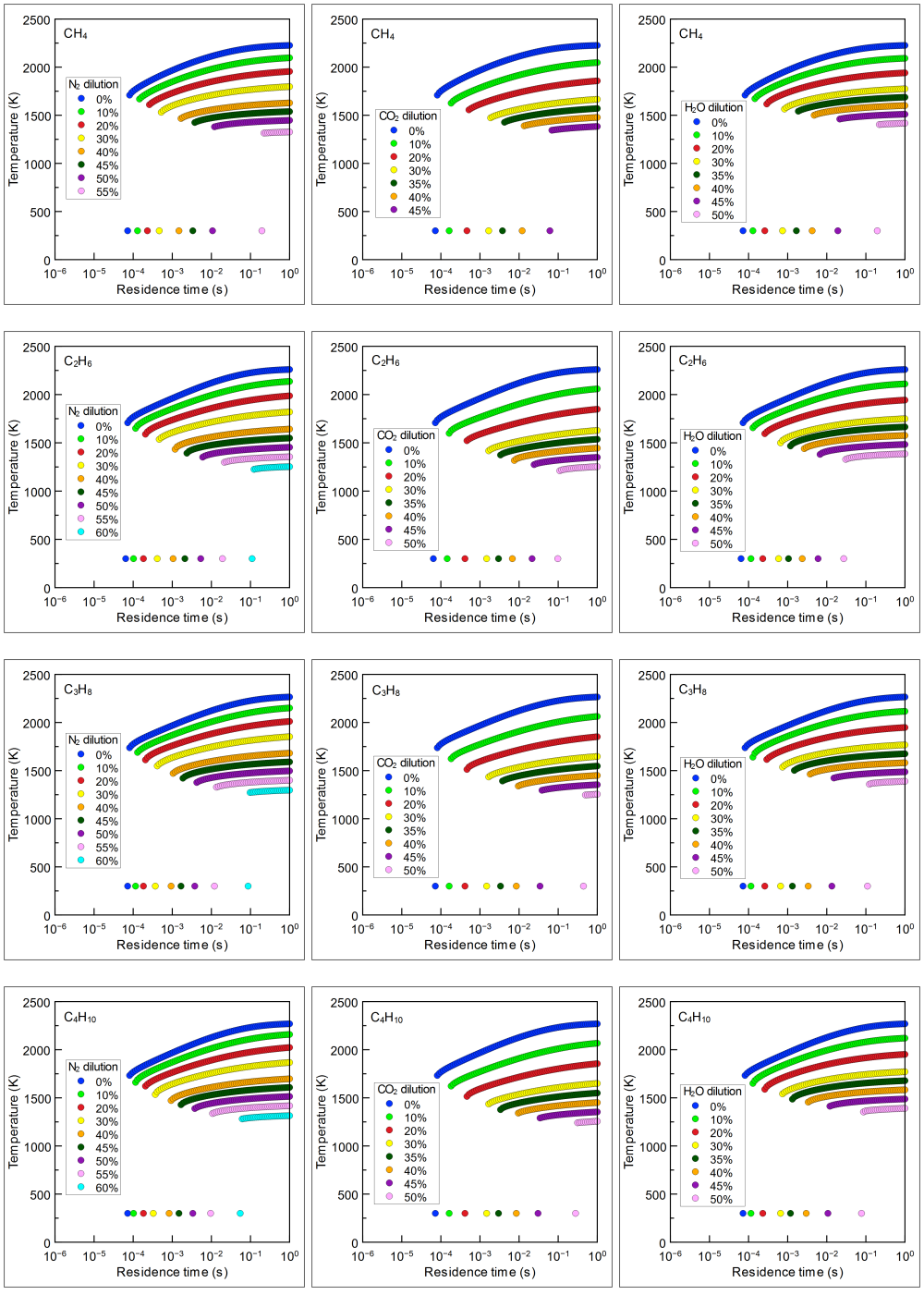


Figure 5: Predicted steady-state temperatures for different fuels as a function of residence time with different types and levels of diluents.

The data for the resulting steady-state temperatures (i.e., critical flame temperatures) versus residence times at extinction have been collected for all the different scenarios. They are plotted in Figure 6. The data for N_2 and CO_2 follow the same trends, which are slightly different than that of H_2O. The differences in the specific heats of the considered diluents, with the one of H_2O being higher on a mass basis (i.e., $c_{p,N_2} = 1.17$ kJ/kg/K, $c_{p,CO_2} = 1.23$ kJ/kg/K and $c_{p,H_2O} = 2.29$ kJ/kg/K [40]), lead to a clear trend namely that the predicted extinction temperatures with H_2O are higher than those of N_2 and CO_2.

The critical flame temperature for a given diluent i, T_{CFT-i}, for the different fuels can be fitted using power law lines in the following form:

$$T_{CFT-i} = c\tau_{mix}^{-d} \tag{5}$$

where c and d are two model parameters that are fuel-dependent. The residence time has been approximated by the mixing time scale, τ_{mix} (ms), which will be taken from the reaction time scale model [41] of FDS in the CFD simulations.

The extinction model parameters, the R-squared coefficient and the range inside which the critical flame temperatures are clipped are reported in Table 3. The considered clipping range is based on typical values reported in the literature for most hydrocarbon fuels [13].

Finally, the critical flame temperature in every cell can be calculated using a mass-weighted average based on the different types of diluents as:

$$T_{CFT} = \sum_i Y_i^{'} T_{CFT-i} \tag{6}$$

where i is the type of diluent (i.e., N_2, CO_2 or H_2O), $Y_i^{'}$ is the normalized mass fraction of diluent i (i.e., $Y_i^{'} = Y_i/(Y_{N2} + Y_{CO_2} + Y_{H_2O})$) and T_{CFT-i} is the critical flame temperature for the type of diluent (i.e., T_{CFT-N_2,CO_2} or T_{CFT-H_2O}). For example, if there is only one diluent in a cell (e.g., N_2, regardless of its mass fraction) the normalized mass fraction of the N_2 diluent will be 1 and only the T_{CFT-N_2,CO_2} curve will be used for estimating T_{CFT}. Note that both T_{CFT-i} curves (i.e., T_{CFT-N_2,CO_2} and T_{CFT-H_2O}) have been constructed considering different diluents and dilution levels, hence, they account for a range of ambient conditions (i.e., ranging from normal air to diluted air). If there are multiple diluents in a cell, then a mass-weighted T_{CFT} value considering one or both T_{CFT-i} curves, depending on the diluents that are present in a cell, will be calculated.

3.3. Discussion

It is worth noting that, there is no distinction between quenched and unquenched fuel in FDS. In reality, it is possible that quenched fuel has undergone

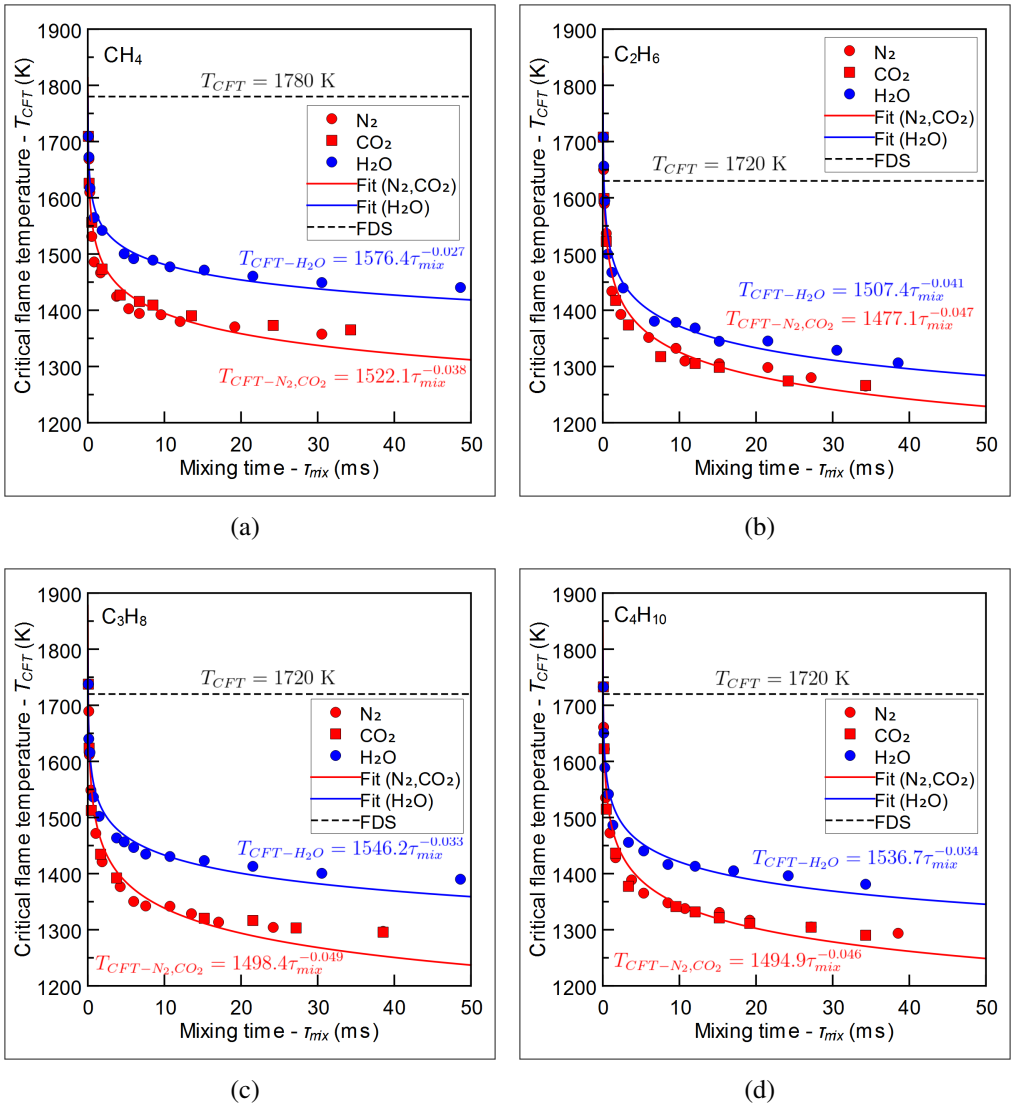


Figure 6: Predicted critical flame temperatures as a function of mixing time for different diluents for (a) CH_4, (b) C_2H_6, (c) C_3H_8 and (d) C_4H_{10}.

some chemistry that changes its re-ignition when it encounters O_2 again. However, this is currently not accounted for in the modelling. The current extinction and re-ignition models available in FDS are deterministic and rely on distinct threshold temperature criteria. The original model used fixed values for T_{ign} and T_{CFT}. Re-ignition was solely on the basis of the cell temperature, and extinction

Table 3: Model parameters used in the extinction model with different fuels.

	c	d	R^2	Clipping range
CH_4				
T_{CFT-N_2,CO_2}	1522.1	0.038	0.96	1200 - 1800 K
T_{CFT-H_2O}	1576.4	0.027	0.98	1200 - 1800 K
C_2H_6				
T_{CFT-N_2,CO_2}	1477.1	0.047	0.98	1200 - 1800 K
T_{CFT-H_2O}	1507.4	0.041	0.98	1200 - 1800 K
C_3H_8				
T_{CFT-N_2,CO_2}	1498.4	0.049	0.97	1200 - 1800 K
T_{CFT-H_2O}	1546.2	0.033	0.95	1200 - 1800 K
C_4H_{10}				
T_{CFT-N_2,CO_2}	1494.9	0.046	0.97	1200 - 1800 K
T_{CFT-H_2O}	1536.7	0.034	0.95	1200 - 1800 K

on the combined basis of cell temperature and mixed species. The new model adds consideration of turbulent mixing to T_{ign} and also of species to T_{CFT}, resulting in computed values for T_{ign} and T_{CFT}. While more physical, the new model still does not account for the prior history of the fuel in the grid cell. For example, fuel exposed to elevated temperatures but low oxygen can undergo chemistry that impacts its future ability to re-ignite and burn. In FDS this cannot currently be accounted for (i.e., there is no ability to distinguish the history of the fuel in a grid cell). In reality, a specific cell temperature, mixing time, and species concentrations may have a range of T_{ign} or T_{CFT} depending on that prior history. Furthermore, the transition between combustion, flame extinction, and potential re-ignition typically can occur across a finite region rather than at distinct threshold temperatures; this behavior is linked to the local thermal state, residence time, heat losses, strain rate, mixing, and dilution levels.

In the current version of the model, the re-ignition temperatures vary with the magnitude of the resolved strain rate in each cell. Considering that flame temperatures, hence the resulting strain rates, might be under-resolved by the CFD grid, the effect of unresolved strain rate could potentially also be incorporated in the model by e.g., using an effective (i.e., resolved and sub-grid scale) strain

rate for determining T_{ign}. The influence of the turbulence model on the model predictions could also be examined in the future. The present study considered the default modelling options of FDS, hence, the effect of different turbulence models was not examined.

In addition, for consistency with the default models used in the code, the residence time from the well-stirred reactor simulations has been approximated with the mixing time scale used by FDS. Further sensitivity study on this choice (e.g., by estimating a true residence time in each CFD cell) and evaluating its impact on the model predictions could be considered in the future.

Overall, the predicted flame temperatures, whether affected by the chosen grid size or the way radiation is modelled, will be important with respect to possible (re-)ignition and extinction of the flame in the simulations. Predicting the correct radiative emission in the simulations without the use of any correction of the radiative fraction, χ_r, particularly if coarse(r) grid sizes are considered, is difficult. As such, under-prediction of the radiative fraction in the simulations can lead to higher flame temperatures, which in turn can affect flame (re-)ignition and extinction. In this case, higher combustion efficiencies are to be expected and full extinction to occur at lower O_2 concentrations. On the other hand, the use of a correction in the radiative emission term, considered by default in engineering type of simulations with FDS, will not be able to accurately capture the reduction in the radiative fraction values typically observed with a decrease of the combustion efficiency. In other words, accurately capturing both the flame temperatures and the radiative fraction in simulations involving flame extinction over a range of grid sizes is quite challenging. A brief sensitivity study on this aspect is presented later for the UMD line burner case.

Finally, the validation cases selected in the paper address gas-phase extinction under controlled fuel supply, and flame extinction is mainly induced by reducing the oxygen concentration in the co-flow. A future next step could include the application of the model in controlled-atmosphere cone calorimeter experiments (see e.g., [52]) where the condensed-phase response is either independently characterized or prescribed. In such scenarios, the proposed new model is expected to affect the predicted gas-phase combustion efficiency and extinction once fuel vapours are released from the solid phase, but it cannot by itself address the oxygen dependence of the condensed-phase degradation process.

4. Experimental cases

Two different experimental scenarios are used for model validation, both of which are part of the MaCFP workshop [19].

The first scenario is the UMD line burner experiments [44], consisting of low-strain, fully-turbulent methane and propane diffusion flames in a canonical line-fire configuration. The burner is centrally located at the outlet of a surrounding oxidizer port (50 cm wide by 75 cm long). The burner is 5 cm wide by 50 cm long with the fuel mass flow rate set to 1 g/s for CH_4 and 1.08 g/s for C_3H_8 (i.e., 50 kW in both cases) while the co-flowing oxidizer (i.e., air) is issued at 85 g/s (total, including variable nitrogen flow). Flame extinction is achieved through dilution via the introduction of gaseous N_2 into the oxidizer. A schematic of the UMD line burner is presented in Figure 7(a).

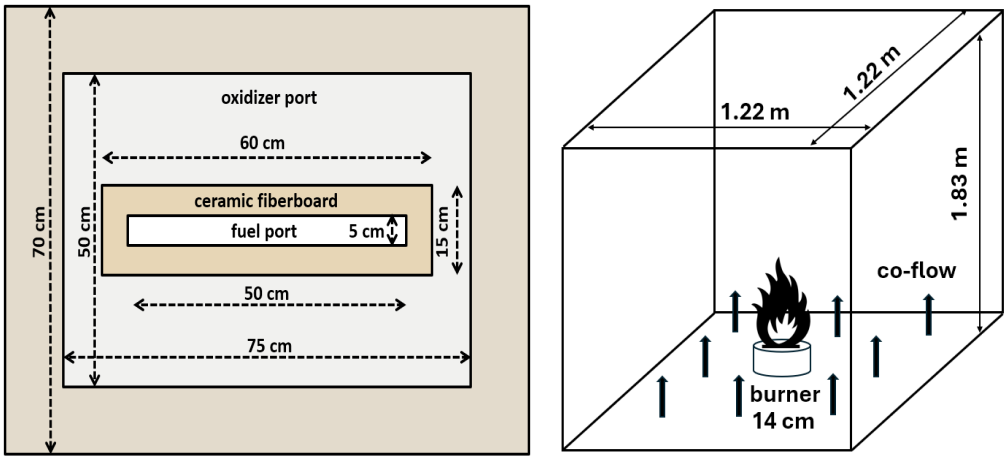
The second scenario is FM burner experiments [45, 46] involving a 14 cm diameter burner with 10 kW methane and propane fires and co-flowing air issued with a velocity of 0.041 m/s. The burner surface is positioned 81 cm below the center of an exhaust hood. Flame extinction is achieved through dilution of the ambient oxygen concentration in the co-flow with N_2 until extinction occurs. A schematic of the FM burner is presented in Figure 7(b).

The combination of these cases provides a sufficient set of experimental data for validation of newly developed extinction models and both have been the focus of many CFD validation studies in the past (e.g., [6, 10, 22, 42, 43]). A more detailed description of the experiments can be found in [47].

5. Numerical setup

FDS version 6.10.1 [1] in LES mode has been used for the numerical simulations presented in the paper. FDS, developed by NIST, is a low-Mach number, second order accurate, finite difference code with explicit time integration. The default FDS sub-models with respect to turbulence (i.e., Deardorff model), combustion (i.e., Eddy Dissipation Concept (EDC)) and radiation (i.e, Gray gas) are considered. A brief overview of these sub-models has been presented in [49] while a detailed presentation of the available models in FDS can be found in [20].

An overview of the numerical setup used in the CFD simulations (i.e., domain, grid size and total number of cells) is presented in Table 4. The finest grid size is chosen a priori based on the D^* criterion by considering values of $D^*/\Delta \geq 15$, which have proven accurate enough for simulations of various fire scenarios with FDS in the past [48]. Based on this, different coarser grid sizes, depending on the



(a) (b)

Figure 7: Overview of the experimental geometry for the (a) UMD line burner and (b) FM burner scenario.

scales involved in each scenario, are also considered as a part of a grid sensitivity study in the simulations.

The ambient temperature and pressure are set to 20°C and 101325 Pa, respectively. Values of 0.24 and 0.33 are assigned for the radiative fraction, χ_r, for CH_4 and C_3H_8, respectively, based on the experimental values reported in [44]. In both scenarios, the O_2 mass fraction in the co-flow remains constant (i.e., 0.23) for the first 5 seconds, until any initial transient effects leave the domain, and then it is reduced by 0.0025/s (by an equivalent increase in the N_2 mass fraction).

The exclusion zone in the UMD line burner case had dimensions 0.6 m x 0.1 m x 0.05 m while in the FM burner scenarios it was 0.16 m x 0.16 m x 0.02 m. The general approach was to include at least one cell surrounding the burners in the transverse direction, to include a cell below the burner surface, and to extend it two cells above the burner in the vertical direction.

In the subsequent sections, the term 'Original model' will refer to the original extinction and re-ignition models of FDS (i.e., relying on constant T_{ign} and T_{CFT} values) while the term 'New model' will refer to the newly proposed models involving a dynamic calculation of T_{ign} and T_{CFT} considering a hot fuel.

Table 4: Overview of the setup used in the numerical simulations.

Case	UMD line burner [44]	FM burner [45, 46]
Fuel	CH_4, C_3H_8	CH_4, C_3H_8
Burner	0.5 m x 0.05 m (line)	0.14 m diameter (circular)
HRR ($\dot{Q}$)	50 kW	10 kW
Domain	1.8 m x 1.6 m x 2 m	1.2 m x 1.2 m x 1.8 m
Grid size (Δ)	0.625 cm[a], 1.25 cm , 2.5 cm	0.5 cm[b], 1 cm, 2 cm
$D^{*,c}$	0.29 m	0.15 m
Resolution index (D^*/Δ)	11.5 - 46	7.5 - 30
Number of cells[d]	5.82 million	4.61 million

[a] Finest grid size in a 1 m x 0.8 m x 1 m region above the burner with a base mesh of 1.25 cm.

[b] Finest grid size in a 0.4 m x 0.4 m x 1.8 m region above the burner with a base mesh of 1 cm.

[c]Characteristic fire diameter [50]: $D^* = \left(\frac{\dot{Q}}{\rho_\infty c_p T_\infty \sqrt{g}}\right)^{2/5}$ where $\dot{Q}$ (kW) is the heat release rate issued from the burner, ρ_∞ (kg/m^3), c_p (kJ/(kg·K)), T_∞ (K) are the density, specific heat and temperature of the ambient air, respectively, while g (m/s^2) is the gravitational acceleration.

[d] Based on the finest grid size, Δ, considered in the simulations.

6. Results

6.1. UMD line burner

A first qualitative comparison of the numerical predictions with the two extinction models is illustrated in Figures 8 - 9, presenting 2D plots of the flame temperature and O_2 concentration for different O_2 concentrations (vol.) in the co-flow with CH_4 as fuel. Similar qualitative results were obtained with C_3H_8 as fuel, not presented here for brevity. Overall, the chosen finest grid size (i.e., 0.625 cm) reveals clear turbulent structures both within the flame and the plume regions for all the cases considered. A reduction in the instantaneous maximum flame temperature within the flame is observed due to dilution of the co-flow with N_2 until

extinction is eventually achieved. This happens faster with 'Original model' compared to 'New model'. At the same time, significantly lower O_2 concentrations are predicted with 'New model' inside the flame at all instances, suggesting higher mixture reactivity and locally fewer extinction events in this region, compared to 'Original model'. This aspect is further discussed later by examining the predicted T_{ign} and T_{CFT} values at distinct axial locations on the centerline of the flames.

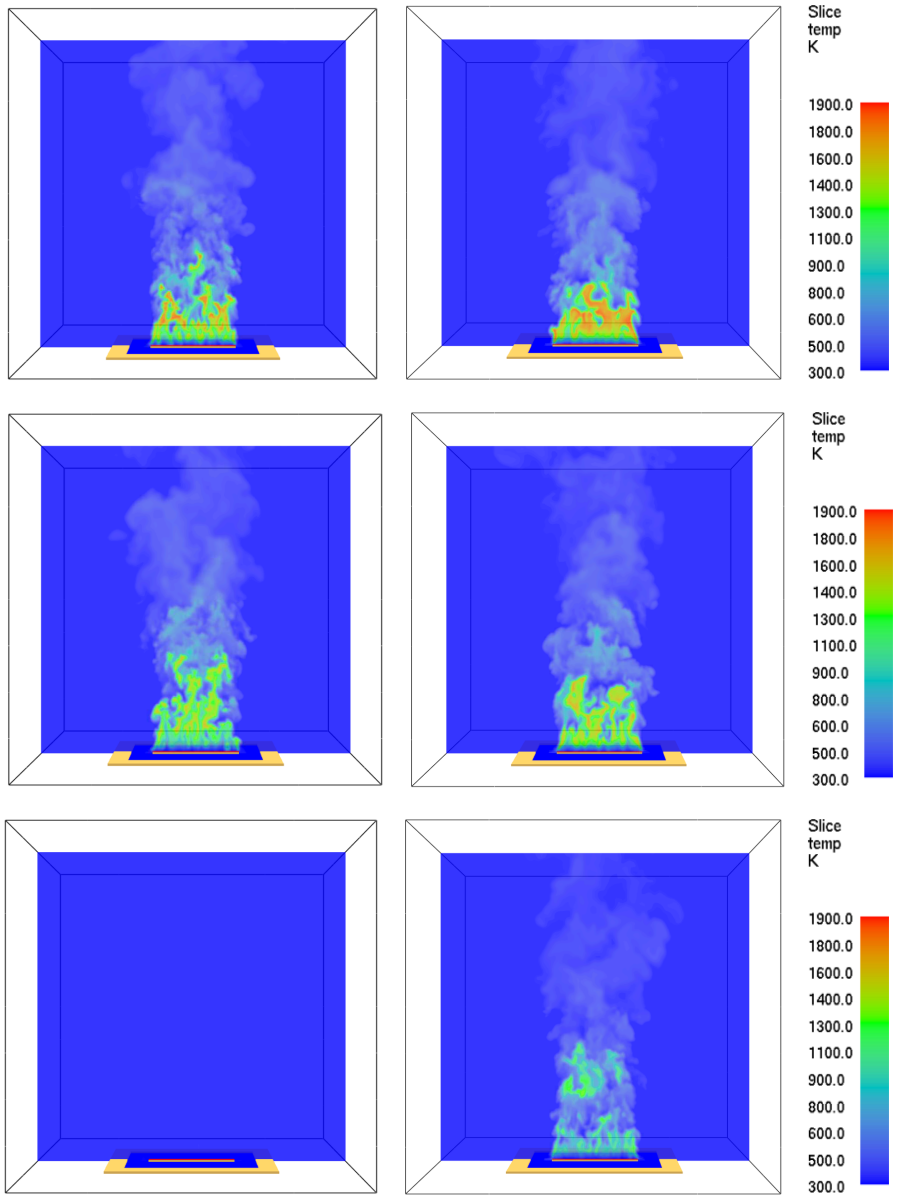
When it comes to the instantaneous predicted values of the strain rate magnitude (see Figure 10), values on the order of 30 - 80 s^{-1} are predicted just above the burner (i.e., near the fuel injection) suggesting a flow transition towards more turbulent conditions with increasing axial distance. The strain rate values increase further downstream and reach values on the order of 80 - 200 s^{-1} in the intermittent region, in line with an increase in the flame temperatures and axial velocities (due to buoyancy). Localized regions of higher strain rates of up to 200 - 300 s^{-1} are observed in high temperature regions. The effect of the local grid refinement strategy (see Table 4), is also visible as the predicted strain rates are significantly higher at heights up to 1 m due to a smaller grid size used in this region.

It is worth noting that the ignition temperatures, T_{ign}, used by 'New model' for determining whether the extinction model will be evaluated or not, are linked to the magnitude of the resolved strain rate in the simulations. When going towards full extinction, the resolved strain rate above the burner decreases (so the T_{ign} values also decrease) and τ_{mix} increases (so the T_{CFT} values also decrease) making it easier for flames to be sustained with 'New model', compared to 'Original model' (which uses constant T_{ign} and T_{CFT} values): full extinction (i.e., $\eta = 0$) is achieved at lower O_2 concentrations.

6.1.1. Combustion efficiencies

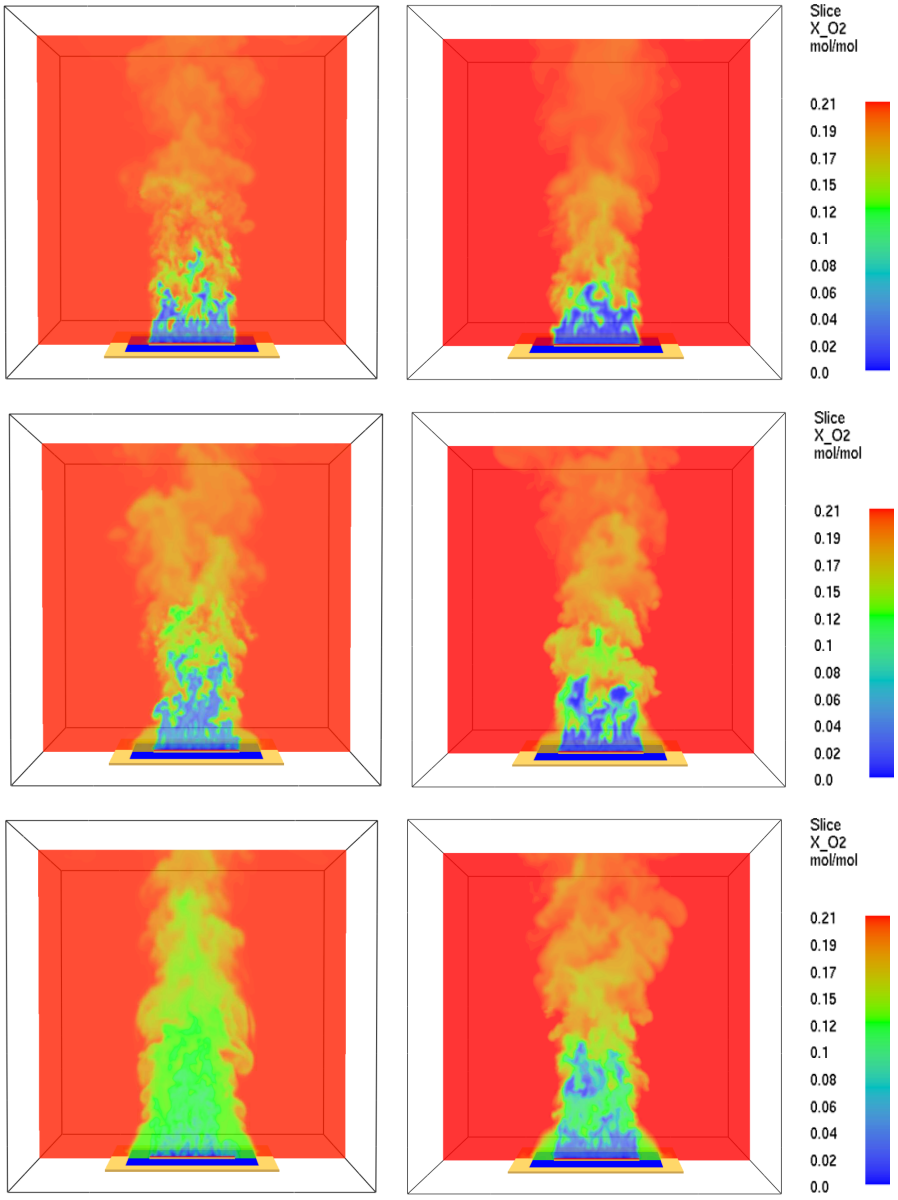
An overview of the predicted combustion efficiencies using CH_4 and C_3H_8 as fuel for the UMD line burner scenarios is presented in Figure 11.

Overall, a different type of grid sensitivity is observed between the predictions of 'Original model' and 'New model'. The former cannot accurately predict the reduction of the combustion efficiency with the use of coarse grids (i.e., 2.5 cm in this case), eventually going to extinction much earlier than expected with both fuels. Under-prediction of the correct flame temperatures inside the flame region when coarse grids are employed in the simulations has a direct influence in the model predictions in this case. When refining the grid size (i.e., 1.25 cm and 0.625 cm), the model predictions improve and exhibit significantly less grid sensitivity, albeit that extinction still occurs earlier compared to the available experimental data and the range of O_2 concentrations from other literature sources.



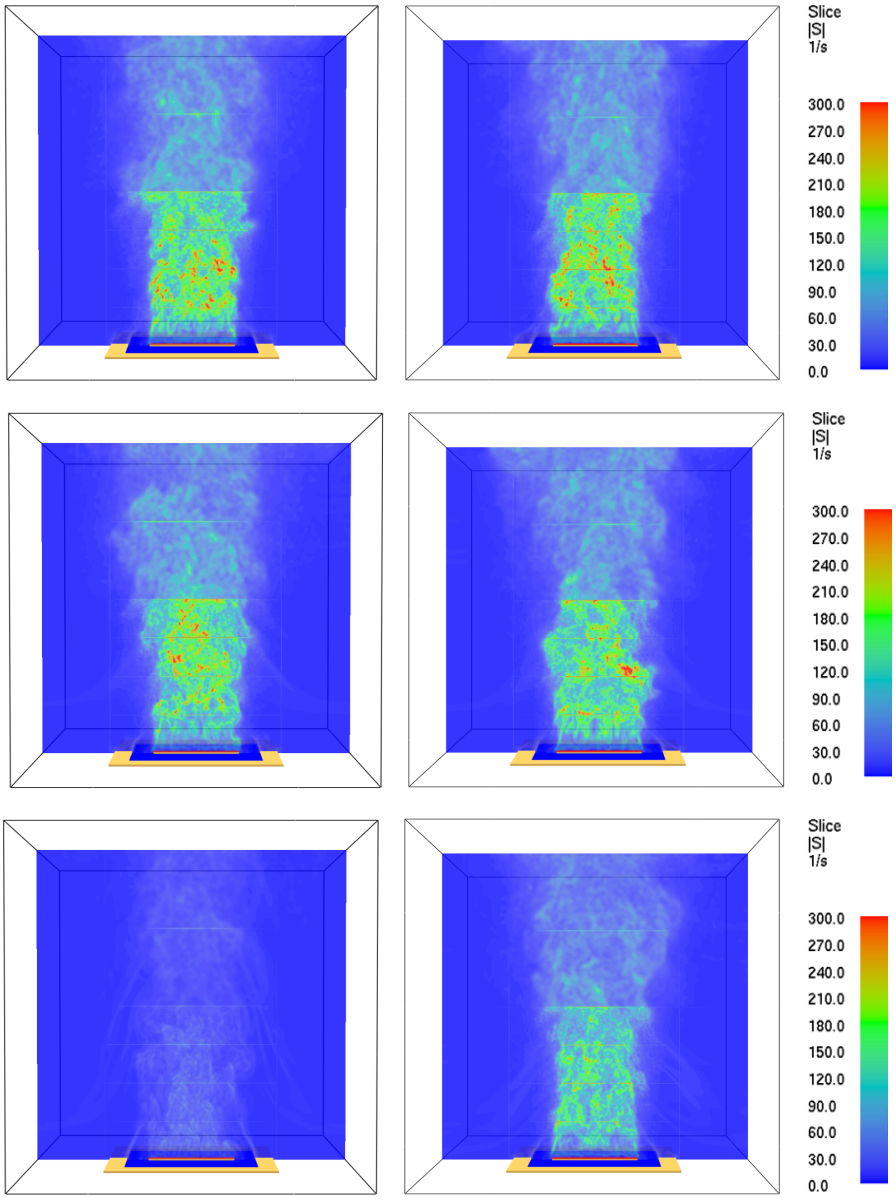
(a) (b)

Figure 8: 2D plots of instantaneous temperature with (a) 'Original model' and (b) 'New model' for an O_2 concentration (vol.) in the co-flow of 21% (top row), 17% (middle row) and 13% (bottom row) with CH_4 as fuel and a grid size of 0.625 cm.



(a) (b)

Figure 9: 2D plots of instantaneous O_2 concentration by volume with (a) 'Original model' and (b) 'New model' for an O_2 concentration (vol.) in the co-flow of 21% (top row), 17% (middle row) and 13% (bottom row) with CH_4 as fuel and a grid size of 0.625 cm.



(a) (b)

Figure 10: 2D plots of instantaneous strain rate magnitude with (a) 'Original model' and (b) 'New model' for an O_2 concentration (vol.) in the co-flow of 21% (top row), 17% (middle row) and 13% (bottom row) with CH_4 as fuel and a grid size of 0.625 cm.

In this case, the combustion efficiency in the beginning of the tests (i.e., when O_2 in the co-flow is still 21% by vol.) is not significantly affected and remains close to 1. A quantitative comparison reporting on the O_2 mole fraction in the co-flow for different reductions in the combustion efficiency using the finest grid size (i.e., 0.625 cm) in the simulations of the UMD line burner scenarios with both models is presented in Table 5.

Table 5: Overview of approximate O_2 mole fraction in the co-flow corresponding to different reductions in the combustion efficiency using the finest grid size (i.e., 0.625 cm) in the simulations of the UMD line burner scenarios.

Combustion efficiency	90%	50%	0%
Fuel: $\mathbf{CH}_4$			
'New model'	0.142	0.131	0.086
'Original model'	0.150	0.145	0.135
Experiment	0.137	0.130	$\approx$0.12
Fuel: $\mathbf{C_3H_8}$			
'New model'	0.135	0.113	0.077
'Original model'	0.139	0.132	0.125
Experiment	0.131	0.127	$\approx$0.09 - 0.1

The model predictions with 'New model' exhibit weaker grid sensitivity in terms of a reduction of the combustion efficiency with all grid sizes, eventually predicting a major reduction in the combustion efficiency around the expected O_2 concentration for both fuels. On the other hand, 'New model' exhibits a stronger grid sensitivity in the beginning of the tests when the O_2 concentration in the co-flow has not yet been significantly reduced. The observed differences between the two models is attributed to the significant variations in the predicted T_{ign} and T_{CFT} values that are used by 'New model' compared to 'Original model'. More specifically, 'New model' will typically require higher T_{ign} values (making re-ignition more difficult) but lower T_{CFT} values (making extinction more difficult) compared to 'Original model' (see results presented later).

It is worth noting that there is a distinct difference in the reduction of the combustion efficiency with C_3H_8 as fuel between the UMD line burner and the FM burner experiments (see the experimental data presented in Figure 11), as opposed to the scenario with CH_4 as fuel, for which the experimental data for both scenarios are quite comparable. Considering that both experiments involved

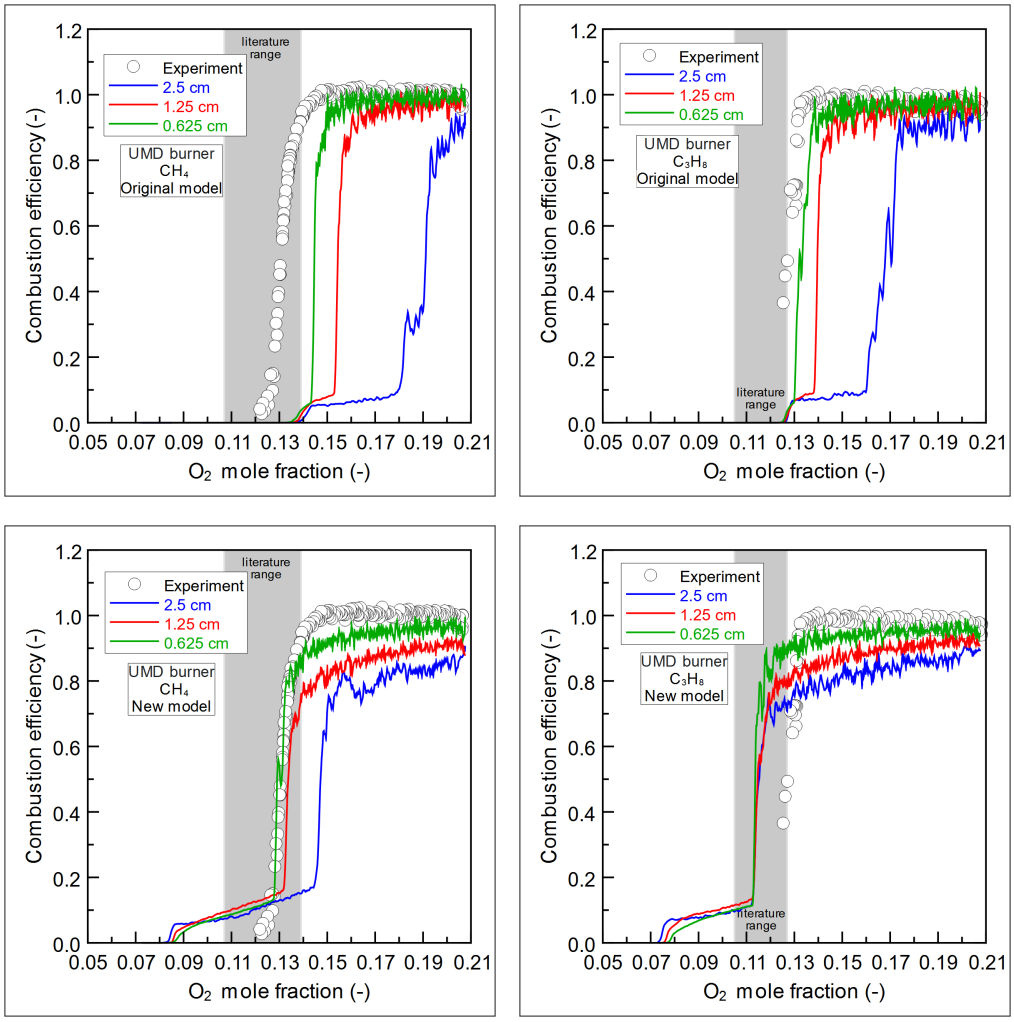


Figure 11: Predicted combustion efficiency in the UMD line burner with (a) CH_4 and (b) C_3H_8 as fuel using 'Original model' (top row) and 'New model' (bottom row).

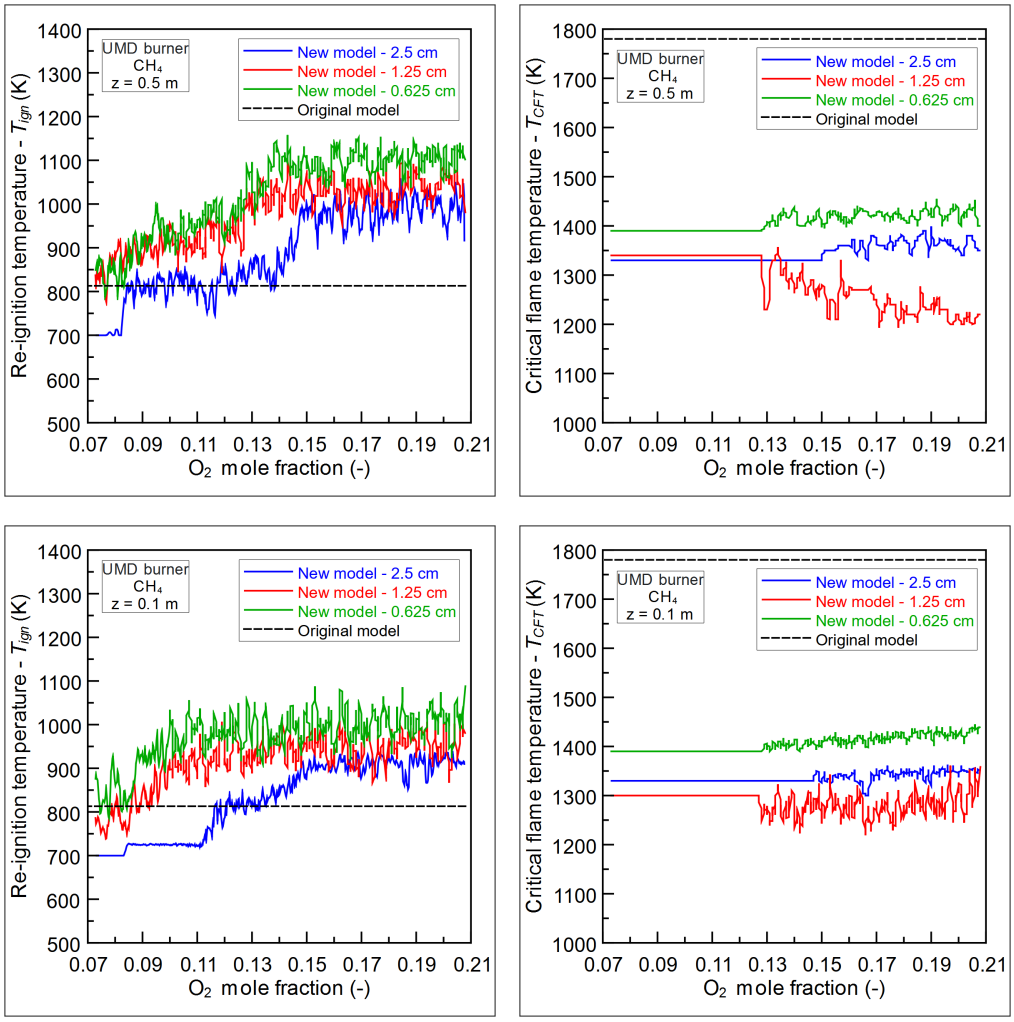
canonical configurations, one would expect a comparable decrease in the combustion efficiency as a function of the O_2 concentration in the co-flow. Despite their variation, both curves are within the reported range of O_2 concentrations for which flame extinction has been reported in the literature in the past.

6.1.2. T_{ign} and T_{CFT}

A more quantitative analysis on the predicted T_{ign} and T_{CFT} values using CH_4 as fuel with 'New model' is presented Figure 12. The constant values used by 'Original model' are also presented in the graphs for comparative purposes. Two locations on the centerline at different axial locations, z, from the burner surface are considered: $z = 0.1$ m, which is representative of the flame region, and $z = 0.5$ m, which is at the end of the intermittent region. The predicted T_{ign} values with 'New model' are usually (considerably) higher than 'Original model'. The T_{ign} values tend to increase with reduced grid size due to better resolution of the flame temperature, which results in higher resolved strain rates as well. At the locations examined, the predicted ignition temperatures vary between 900 - 1150 K, depending on the grid size, in the beginning of the tests when the co-flow is not yet strongly diluted. This implies that (re-)ignition with 'New model' is more difficult compared to 'Original model' and, as such, the primary extinction model based on the enthalpy balance will not be evaluated at all in many cells (i.e., combustion will be suppressed automatically). This also explains the quantitative differences in the predicted combustion efficiencies observed between the two models in the beginning of the tests in Figure 11. A reduction of the O_2 concentration in the co-flow slowly leads to flame extinction, lowers the flame temperature and the resolved strain rate which eventually also leads to a decrease of the T_{ign} values, postponing full extinction when compared to the 'Original model'.

The predicted T_{CFT} values remain in the range of 1200 -1450 K within the flame region and exhibit some grid sensitivity. The variation of the predicted TCF T values with respect to grid is partially linked to the mixing time scale as implemented in FDS: different time scales become dominant on different grid sizes. Once extinction occurs, the T_{CFT} values tend to a constant value, fixed by the mixing time scale model. Overall, it is clearly evident that the critical flame temperatures predicted with 'New model' are significantly lower than the constant values assigned to the 'Original model' (i.e., 1780 K for CH_4).

If ignition inside a cell occurs with both models, it will be more difficult for extinction to happen with 'New model'. On the other hand, (re-)ignition inside a cell will be easier to occur with 'Original model'. As such, there are some distinct qualitative / quantitative differences occurring between the two sub-models for (re-)ignition and extinction with 'New model' and 'Original model'. The constant values assigned with 'Original model' correspond to nearly-zero strain rates for the determination of T_{ign} and very small mixing times (i.e., small residence times)



(a) (b)

Figure 12: Predicted (a) re-ignition temperatures, T_{ign}, and (b) critical flame temperatures, T_{CFT}, in the UMD line burner with CH_4 as fuel on the centerline at different axial locations ,z, from the burner using 'New model'.

for the determination of T_{CFT}.

6.1.3. Sensitivity study

A sensitivity study on the influence of the size of the exclusion zone (EZ) is presented with 'Original model' and 'New model' in Figure 13. The default

height of the exclusion zone above the burner (i.e., 0.05 m) was doubled (i.e., 0.1 m) while the other dimensions (see Section 5) remained unchanged. It can be observed that the size of the exclusion zone has a stronger influence in the predicted combustion efficiencies on the coarse(r) grid size simulations (i.e., the scenarios where the reaction rates are typically found very close to the burner surface). As the grid is refined, the reactions typically shift further downstream, thus reducing the dependency of the numerical predictions on the size of the exclusion zone. As expected, extinction is obtained at lower O_2 concentrations for larger EZs as now combustion is sustained in a larger zone above the burner. A similar behavior is observed for both fuels considered in the simulations (i.e., CH_4 and C_3H_8) and for both models tested.

An overview of the predicted combustion efficiencies using CH_4 and C_3H_8 as fuel for the UMD line burner scenarios without a correction in the radiative emission term (i.e., no χ_r correction) is presented in Figure 14. The reader is referred to [50, 51] for the details on how this χ_r correction is considered in the modelling with FDS. To complement the results, the predicted radiative fractions from the same simulations are also presented in Figure 15.

Overall, the radiative fractions are under-predicted, regardless of the model employed in the simulations, by up to 50 - 60$\%$ in some cases. As expected, the predicted χ_r values increase with decreasing grid cell size due to better resolution of the flame temperatures, which increase the radiative emission of the flame. Some spikes in the predicted radiative fractions are occasionally predicted just before full extinction occurs. These are a numerical artefact. In general, under-prediction of the radiative fractions in the simulations leads to higher flame temperatures which in turn results in typically higher combustion efficiencies and full extinction occurring at lower O_2 concentrations in the co-flow when compared to the same cases with the χ_r correction enabled.

6.2. FM burner

An overview of the predicted combustion efficiencies with CH_4 and C_3H_8 as fuel for the FM burner case scenarios is presented in Figure 16. Overall, the observations previously made for the UMD line burner case are confirmed. The temperature fields, oxygen concentrations and strain rates are not reported for brevity, since they provide very similar conclusions to the UMD burner case. In general, a relatively finer grid resolution (i.e., 0.5 cm in this case) is needed in order to get the correct combustion efficiency in the beginning of the experiments, where no dilution of the co-flow has yet occurred, with either 'Original model' or 'New model'. It is worth noting that the coarser grid resolutions considered here

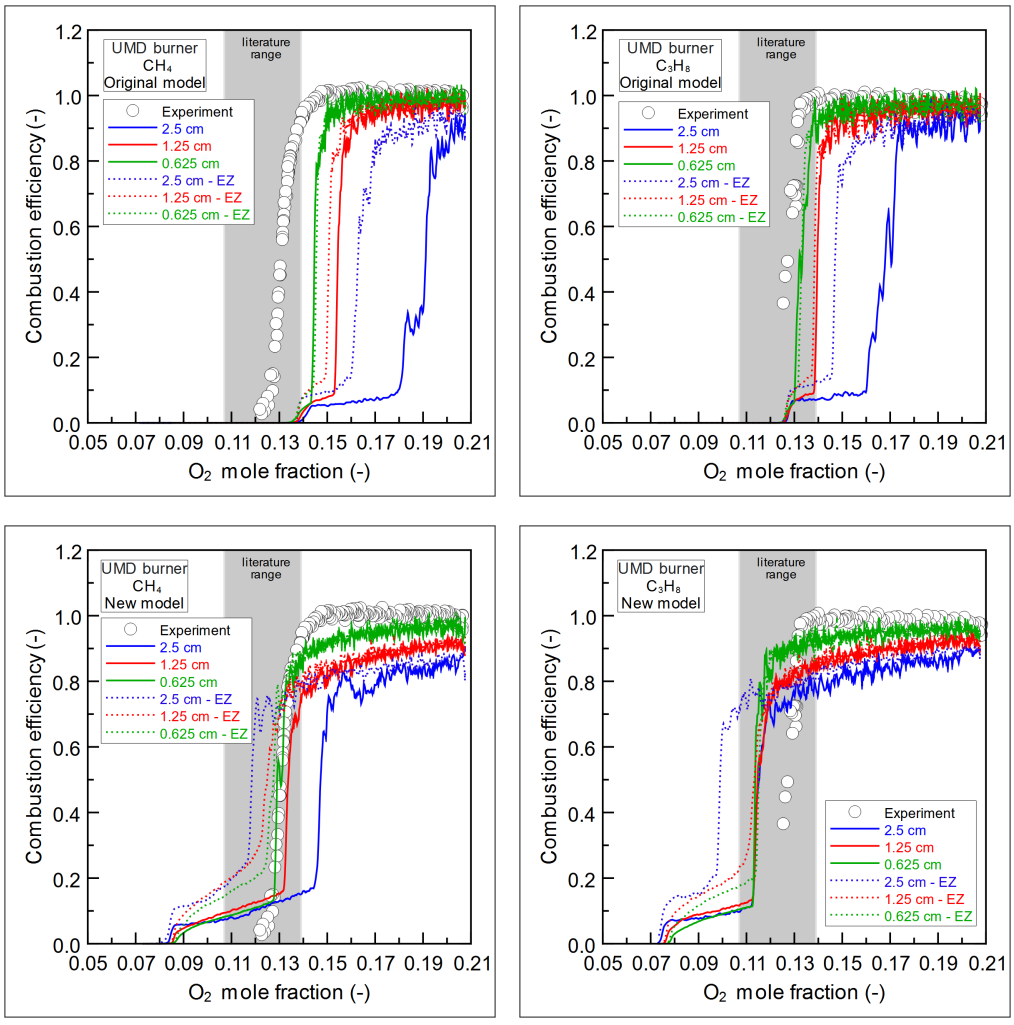


Figure 13: Predicted combustion efficiency in the UMD line burner with (a) CH_4 and (b) C_3H_8 as fuel using 'Original model' (top row) and 'New model' (bottom row) using the default and a larger exclusion zone (EZ).

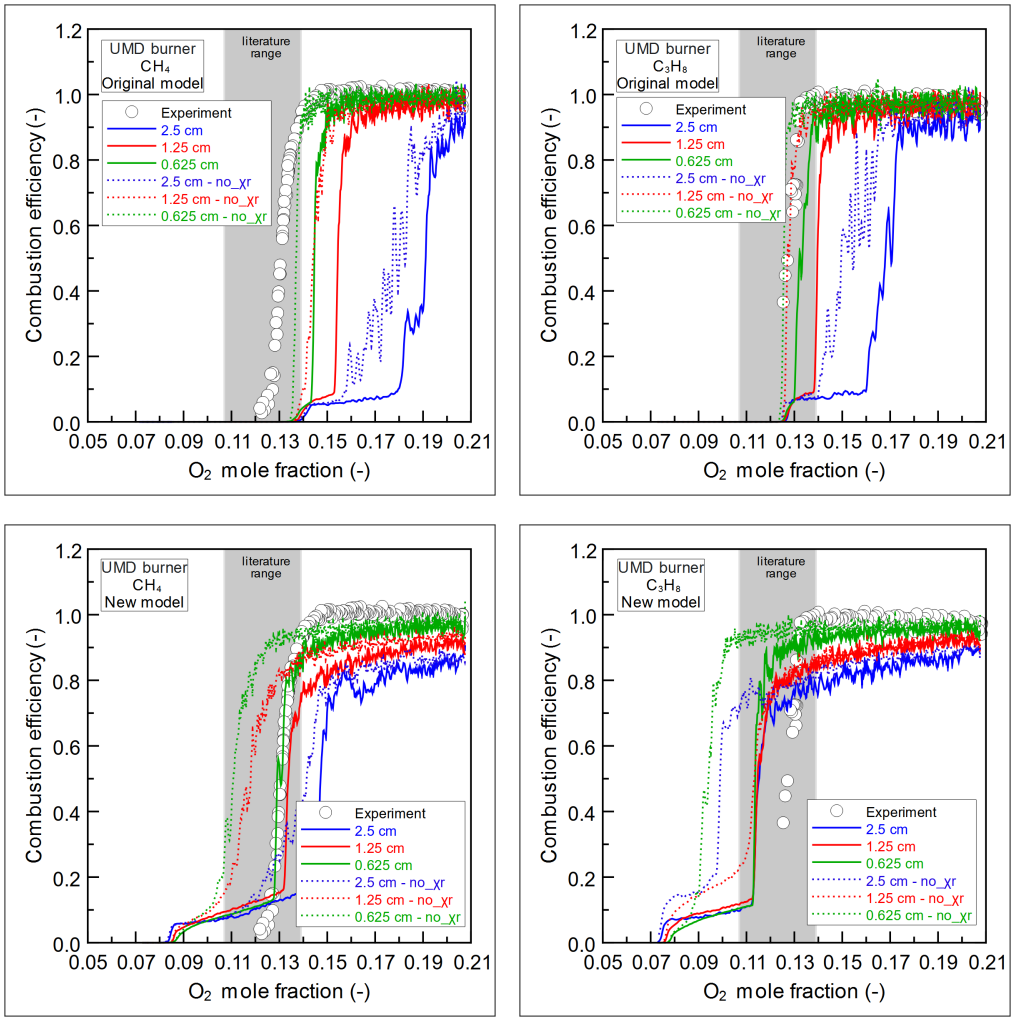


Figure 14: Predicted combustion efficiency in the UMD line burner with (a) CH_4 and (b) C_3H_8 as fuel using 'Original model' (top row) and 'New model' (middle row) without χ_r correction.

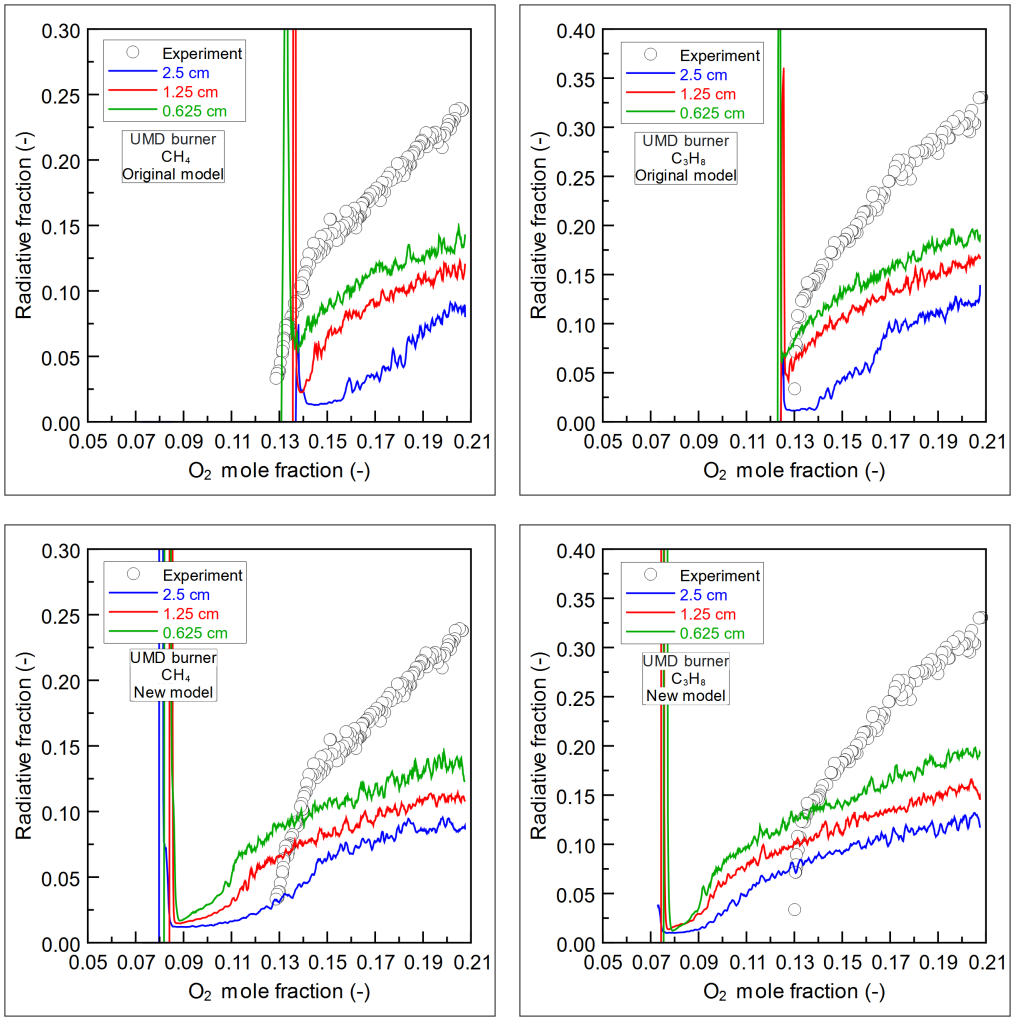


Figure 15: Predicted radiative fraction in the UMD line burner with (a) CH_4 and (b) C_3H_8 as fuel using 'Original model' (top row) and 'New model' (middle row) without χ_r correction.

(i.e., 2 cm and 1 cm) only resulted in 7 and 14 cells across the burner diameter, respectively, with only the latter grid size being somewhat sufficient for resolving the flame plume dynamics. The predicted combustion efficiency during these early stages of the tests is more negatively impacted in this case with 'New model' due to the higher T_{ign} values required compared to 'Original model'. Again, the major reduction in the combustion efficiency is fairly well predicted with 'New model' even though the full extinction is achieved later compared to the experiments due to the reasons discussed earlier. On the other hand, the major decrease in the combustion efficiency with 'Original model' occurs earlier compared to the experiments for both fuels considered. A quantitative comparison reporting on the O_2 mole fraction in the co-flow corresponding to different levels of combustion efficiency using the finest grid size (i.e., 0.5 cm) in the simulations of the FM burner scenarios with both models is presented in Table 6.

Table 6: Overview of approximate O_2 mole fraction in the co-flow corresponding to different reductions in the combustion efficiency using the finest grid size (i.e., 0.5 cm) in the simulations of the FM burner scenarios.

Combustion efficiency	90%	50%	0%
Fuel: $\mathbf{CH_4}$			
'New model'	0.139	0.132	0.09
'Original model'	0.152	0.149	0.132
Experiment	0.135	0.124	≈0.12
Fuel: $\mathbf{C_3H_8}$			
'New model'	0.136	0.123	0.08
'Original model'	0.147	0.135	0.122
Experiment	0.155	0.108	≈0.10

It should be stressed that the HRR of the fires in the FM burner scenarios is rather small (i.e., 10 kW) and significantly larger fires are typically considered in real life / engineering type of scenarios. Something similar can be said about the UMD line burner scenario (i.e., 50 kW), which posed limitations in the finest grid size considered in the simulations given the nature of the burner configuration (i.e., line). As such, the observed grid dependency with either model will be greatly diminished for larger fires, which are typically much better resolved (i.e., larger D^*/Δ values).

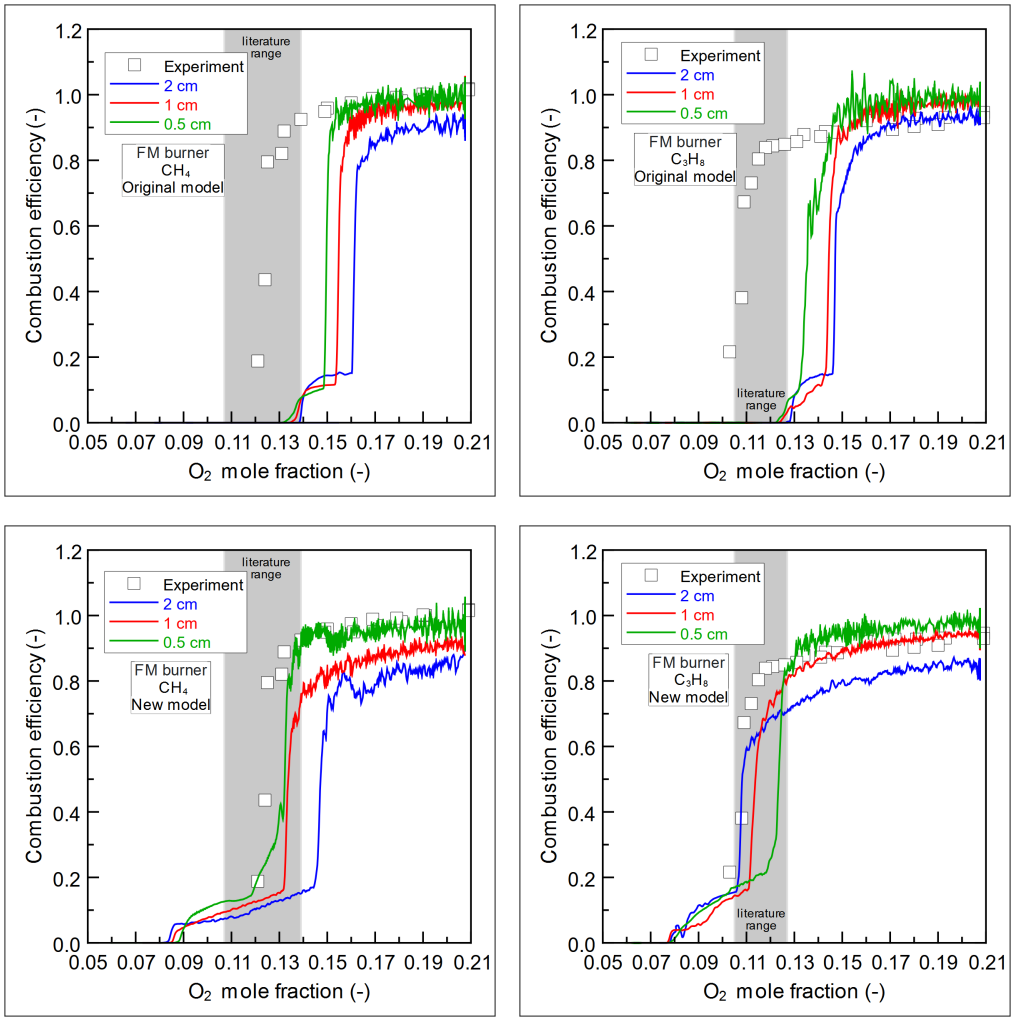


Figure 16: Predicted combustion efficiency in the FM burner with (a) CH_4 and (b) C_3H_8 as fuel using 'Original model' (top row) and 'New model' (bottom row).

7. Conclusions

The research study presented the development of a new dynamic flame extinction and re-ignition model that considers the local properties of the flow. As such, flame extinction was modelled considering an enthalpy balance based on the concept of a critical flame temperature, T_{CFT}. Re-ignition was allowed if the local

cell temperature was above a threshold re-ignition temperature, T_{ign}. Within the new model, T_{CFT} was a function of the local mixing time and species concentrations and was estimated from well-stirred reactor simulations considering different diluent concentrations. The T_{ign} values were a function of the local strain rate magnitude and were derived from counterflow simulations of hot fuel and oxidizers. The model was implemented in the Fire Dynamics Simulator (FDS) version 6.10.1 and was evaluated in LES mode. Model implementation consisted of a new user input to invoke the model, editing the T_{ign} subroutine to use the new approach when requested, and the addition of a new subroutine for the calculation of T_{CFT} and extinction. Comparisons to the original extinction and re-ignition models of FDS, which employ constant T_{CFT} and T_{ign} values, were made. For validation purposes of the new model, the UMD line burner and FM burner scenarios, both part of the MaCFP workshop [19, 47], were selected.

Overall, the performance of the new dynamic model was very satisfactory across all fuels and scenarios tested and was able to correctly capture the decrease of the combustion efficiency provided the grid resolution was satisfactory. The use of a coarse(r) grid affected the predicted combustion efficiency of the model in the early stages of the tests (i.e., when dilution slowly occurs). However, it did not affect the onset of the bulk reduction of the combustion efficiency (i.e., when going towards full extinction).

In general, the predicted T_{ign} values were typically higher than the constant values employed by FDS at the various centerline locations above the burners. On the other hand, the predicted T_{CFT} values were significantly lower than the constant values employed by FDS (e.g., 1200 - 1450 K with the new model versus 1780 K for the original model with CH_4 as fuel).

The model showed great potential in capturing flame extinction under different scenarios involving different fuels. Validation of the model involving ventilation-limited compartment fire scenarios with variable heat release rates has also been successfully performed [49]. Further model extension by considering additional fuels and model validation in scenarios involving transition from fuel-controlled to ventilation-controlled compartment fires with external flaming is planned in the future.

Acknowledgments

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